

Pulse 1.1.0

QA Automation & Test Report

CROSS-BROWSER SUITE — 628 / 637 PASSED

4 flaky (recovered) · 1 data-timing fail · 4 skipped · 22 defects logged separately

Application: Pulse 1.1.0 — <https://test.alt-pulse.com>

Build: v1.1.0_TEST · **Browsers:** Chrome, Edge, Firefox, Brave

Framework: Playwright + live Playwright MCP · **Date:** 2026-06-18

Pulse 1.1.0 — Consolidated Test Report

Application: Pulse 1.1.0 — Building Operations Platform (<https://test.alt-pulse.com>)

Build marker: `v1.1.0_TEST` · © 2026

Account under test: Organization Admin (ALEC organization) + a scoped Facility Manager sub-user

Test framework: Playwright `@playwright/test` ^1.61.0 (TypeScript) + live Playwright MCP exploratory pass

Report date: 2026-06-18

This is the **all-in-one summary**. Companion documents: the **Defect Report** ([Bug_Report_Pulse_1.1.0.pdf](#)) lists every defect with reproduction steps and screenshots, and the **Comprehensive Test Case** document ([Pulse_1.1.0_Comprehensive_Test_Cases.md](#)) holds the full manual case catalogue.

1. Verdict at a glance

Metric	Value
Automated test executions	637
Passed	628 (98.6 %)
Flaky (passed on retry)	4
Failed (final)	1 — live-data first-paint timing, not an app defect
Skipped	4 — conditional/known-issue guards
Browsers covered	Chrome, Edge, Firefox, Brave
Testing types covered	9 (functional, UI/UX, cross-browser, responsive, security, performance, RBAC, exploratory, smoke/auth)
Wall-clock (full combined run)	18.4 min

Two-track verdict. The **automated cross-browser suite is functionally green** — the single failure and four flakes are all live-data / first-paint latency on the staging server and pass on isolated re-run. A separate **live exploratory + RBAC pass** surfaced **22 defects** (6 High · 8 Medium · 8 Low), all catalogued in the Defect Report. The two High-priority items are RBAC/security issues (broken access control + dashboard data-scope leak).

2. Totals by suite

Each cross-browser suite runs on all four browsers (Chrome / Edge / Firefox / Brave). Resolution and accessibility-style layout checks are engine-independent and run on one engine.

#	Suite (folder)	What it checks	Unique tests	Projects
1	Auth	Login, credentials, injection/XSS safety, route protection	12	anonymous (Chrome)
2	Core	Portfolio Dashboard, Asset Management, Overview Map	17	4 desktop
3	AI & Intelligence	AI Chat, Insights, Reports, Energy, Compliance	14	4 desktop
4	AI Engineer	HVAC Optimization, Smart Commissioning	6	4 desktop
5	Operations	Work Orders, Maintenance Calendar, Fault Detection	9	4 desktop
6	Settings	Profile, Theme, Users, Roles, Permission Matrix, Workflow	29	4 desktop
7	Shell	Navigation, UI consistency, responsive layout	9	4 desktop
8	Quality	Rendering, performance budget, known-issue guard	5	Chrome

Total unique automated test cases catalogued: 101. The cross-browser suites execute on 4 browsers (plus tablet/mobile/performance projects), giving **637** total executions in the combined run.

3. What kind of testing was done

Testing type	Covered?	How
Functional (features work)	✔ Yes	Positive, negative & edge cases across every module (Core, AI, AI-Engineer, Operations, Settings)
UI / UX (looks right)	✔ Yes	Fonts, dark-theme palette, alignment, layout, no horizontal page overflow, no clipping
Cross-browser	✔ Yes	Chrome, Edge, Firefox, Brave
Resolution / responsive	✔ Yes	Desktop 1440×900 and mobile 390×844 audited per page; responsive project on tablet + mobile
Security	✔ Yes	HTTPS, token storage, password leakage, unauthenticated route redirects, SQL-injection / XSS inputs, RBAC route protection
Performance	✔ Yes	Page-load budget (≤ 20 s), interactivity after navigation, console-error flood guard
RBAC (role-based access)	✔ Yes	Role creation + permission matrix + workflow gating; live sub-user scope enforcement
Smoke / Auth	✔ Yes	App reachability, login success/failure, session handling, logout
Accessibility (automated)	⚠ Partial	Landmark / heading / <code>h1</code> - <code>main</code> structure captured in the UI audit; dedicated axe-core scan not yet wired for 1.1.0
Load / stress	✘ No (gap)	Concurrent-user / soak simulation out of scope — see §8

4. Browser coverage

Browser	Engine	Project	Status
Chrome	Chromium	<code>chrome</code> (channel: chrome)	✓ Pass
Edge	Chromium	<code>edge</code> (channel: msedge)	✓ Pass
Firefox	Gecko	<code>firefox</code> (bundled)	✓ Pass
Brave	Chromium	<code>brave</code> (executablePath + hardening flags)	✓ Pass

Supporting projects: `setup` (one-time authentication), `anonymous` (logged-out security/redirect flows), `performance` (Chrome), `tablet` (iPad gen 7 landscape), `mobile` (Pixel 7).

5. Feature coverage by module

Core — Portfolio Dashboard (hero, hierarchy counters, Energy/Water/Cost/CO₂ KPIs, alarm summary, building-performance cards, immediate-actions panel), Asset Management (BMS section, inventory summary, device categories, site scope), Overview Map (map canvas, 2D/3D toggle, site list, hover search).

AI & Intelligence — AI Chat (composer, message echo, response), AI Insights (hierarchy selector, contextual insights, recommended actions), AI Reports (header + seven report tabs, section switching, placeholder guard), Energy Intelligence (opportunity filters), Compliance (frameworks, gap analysis, roadmap, AI advisor).

AI Engineer — HVAC Optimization and Smart Commissioning (Integration-Pending marker, content render, graceful no-crash).

Operations — Work Orders, Maintenance Calendar, Fault Detection (Integration-Pending marker, content, graceful no-crash).

Settings — Profile (account details, access level, activity), Theme Settings (light/dark modes, palette tokens, import/export/reset/save), Users (table, stat cards, status tabs, search, pagination, 3-step invite wizard with validation), Roles (create-role wizard, status filters), Permission Matrix (MODULE column + role columns), Workflow (role hierarchy nodes, zoom/fit controls).

Shell & Quality — global navigation (every route loads, reload stability, sidebar rail, logo home), UI consistency (no overflow, fonts, footer version, theme toggle), rendering (integrated pages show real content; not-integrated pages fail gracefully), performance budget, known-issue guard.

Auth (anonymous) — login element rendering, password masking, valid/invalid credential handling, empty/malformed submissions, SQL-injection & XSS safety, unauthenticated deep-link redirects, full login→logout flow.

6. Security & RBAC highlights

Anonymous security (passing): all protected routes redirect unauthenticated visitors to login; SQL-injection and XSS payloads in the login form do not bypass auth or execute; passwords are masked.

RBAC configuration (passing): the create-role wizard, granular per-module permission matrix (view / add / edit / delete / download / control), and **workflow gating** (an unassigned role is not selectable for a user until placed in a workflow) all behaved correctly.

RBAC enforcement (live sub-user — mixed): a Facility Manager scoped to **Marina Plaza** was created and driven live. Navigation, Overview Map, Asset Management and Reports correctly scoped to the single site. However two **High-severity** gaps were found and logged in the Defect Report:

- **BUG-020 — Broken access control:** admin pages (`/settings/users` , `/settings/roles` , `/settings/roles/permission-matrix`) are reachable by the non-admin via **direct URL**; route guards are missing.
 - **BUG-021 — Dashboard data-scope leak:** the Portfolio Dashboard shows full-organization Devices/Water/Cost/CO₂ totals to the site-scoped user, while Asset Management correctly scopes to 552 devices.
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7. Known / expected states (excluded from defects)

Per the development team, the following are **known** and **excluded** from the defect count:

1. The hero data-mode switcher labels "7 Days" as "8 Days".
 2. Multi-organization support is still in progress.
 3. Responsive design is still in progress (mobile reflow incomplete on some tables).
 4. AI Insights / Recommendations use Building-level data for Site/Org views.
 5. Six modules are **not yet integrated** and intentionally show an "Integration Pending" placeholder: `/smart-cx`, `/start-stop`, `/maintenance`, `/maintenance-calendar`, `/fdd`, `/compliance`. All six were verified to render the placeholder cleanly with **no crash**.
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8. Honest gaps (out of scope)

- **Load / stress testing** — no concurrent-user / soak simulation was performed.
 - **Full automated accessibility (axe-core)** — a dedicated WCAG 2.1 A/AA scan is not yet wired into the 1.1.0 suite; landmark/heading structure was captured via the UI audit only.
 - **API / backend testing** — scope is the web UI against live staging; backend endpoints were observed (via network capture) but not directly contract-tested.
 - **Live-data flakes** — the suite runs against live staging, so a few specs are sensitive to first-paint latency; CI retries absorb these.
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9. How to run

PowerShell on this machine blocks the `npx.ps1` shim, so the suite is invoked through Node directly:

```
# Full combined run - all browsers + projects
node "node_modules/@playwright/test/cli.js" test --reporter=list,html,json

# Single browser
node "node_modules/@playwright/test/cli.js" test --project=chrome

# Re-run only failures (absorbs transient live-data flakes)
node "node_modules/@playwright/test/cli.js" test --last-failed

# Open the interactive HTML report
node "node_modules/@playwright/test/cli.js" show-report
```

10. Artifacts

- **Interactive HTML report:** `playwright-report/index.html`
 - **Machine-readable results:** `test-results/results.json` · run log `test-results/run-log.txt`
 - **Failure evidence:** screenshots / video / trace auto-captured under `test-results/<test>/` on failure (retain-on-failure).
 - **Defect Report (with screenshots):** [QA/bug-report/Bug_Report_Pulse_1.1.0.pdf](#)
 - **Live exploratory screenshots:** [qa-live/](#) (33 captures) and audit screenshots [QA/bug-report/screenshots/](#) (38 captures).
 - **Comprehensive manual test cases:** [QA/Pulse_1.1.0_Comprehensive_Test_Cases.md](#)
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Appendix A — Complete automated test-case inventory

Every automated test title, grouped by spec file, auto-extracted from the test sources. Cross-browser suites execute on 4 browsers; parameterised titles (route loops) expand to one execution per route at run time.

tests/e2e/specs/ai-engineer/hvac-optimization.spec.ts — 3 tests

#	Test
1	shows the Integration Pending marker
2	renders today's conditions and algorithm intelligence
3	loads without crashing or leaking framework errors

tests/e2e/specs/ai-engineer/smart-commissioning.spec.ts — 3 tests

#	Test
1	shows the Integration Pending marker
2	renders commissioning phases and savings impact
3	loads without crashing or leaking framework errors

tests/e2e/specs/ai/ai-chat.spec.ts — 2 tests

#	Test
1	renders the chat composer
2	sending a message echoes it into the conversation

tests/e2e/specs/ai/ai-insights.spec.ts — 3 tests

#	Test
1	renders the hierarchy selector
2	shows Contextual Insights and AI Recommended Actions sections
3	renders substantive content with no leaked placeholders

tests/e2e/specs/ai/ai-reports.spec.ts — 4 tests

#	Test
1	renders the report header and hierarchy selectors
2	exposes all seven report section tabs
3	switching to the Energy section keeps content populated
4	no leaked placeholder values in the report

tests/e2e/specs/ai/compliance.spec.ts — 3 tests

#	Test
1	shows the supported certification frameworks
2	renders gap analysis, roadmap and AI advisor sections
3	stays inside the authenticated shell without crashing

tests/e2e/specs/ai/energy-intelligence.spec.ts — 2 tests

#	Test
1	renders opportunity category filters
2	switching a category filter keeps the list populated

tests/e2e/specs/auth/login.anon.spec.ts — 10 tests

#	Test
1	renders all login elements correctly
2	password input masks the entered value
3	Forgot Password link navigates to recovery page
4	positive: valid credentials log the user in
5	negative: wrong password keeps the user on login
6	negative: unknown account is rejected
7	negative: empty submission does not authenticate
8	negative: malformed email is not accepted
9	security: SQL injection in credentials does not bypass auth
10	security: XSS payload in email field is not executed

tests/e2e/specs/auth/route-protection.anon.spec.ts — 2 tests

#	Test
1	security: ... redirects an unauthenticated visitor to login
2	full login then logout flow

tests/e2e/specs/core/asset-management.spec.ts — 4 tests

#	Test
1	renders the page heading and organization scope
2	renders the BMS section and asset inventory summary
3	lists the portfolio sites and any loaded device categories
4	shows substantive catalog content with no leaked placeholders

tests/e2e/specs/core/overview-map.spec.ts — 5 tests

#	Test
1	renders the map canvas
2	exposes the 2D/3D view toggle
3	shows the site list and a (hover-revealed) search box
4	lists the portfolio sites
5	no leaked placeholder values in the site list

tests/e2e/specs/core/portfolio-dashboard.spec.ts — 8 tests

#	Test
1	shows the personalised hero with a Live status

#	Test
2	hierarchy counters (Sites/Buildings/Floors/Devices) render numbers
3	KPI tiles for Energy, Water, Cost and CO ₂ are present with units
4	alarm summary shows total and severity breakdown
5	Building Performance section lists building score cards
6	"view full report" link opens the reports page
7	Immediate Actions Required panel is rendered
8	dashboard contains no leaked placeholder values

tests/e2e/specs/operations/fault-detection.spec.ts — 3 tests

#	Test
1	shows the Integration Pending marker
2	renders diagnosis and impact-summary sections
3	loads without crashing or leaking framework errors

tests/e2e/specs/operations/maintenance-calendar.spec.ts — 3 tests

#	Test
1	shows the Integration Pending marker
2	renders a month calendar view
3	loads without crashing or leaking framework errors

tests/e2e/specs/operations/work-orders.spec.ts — 3 tests

#	Test
1	shows the Integration Pending marker
2	renders work-order KPI content
3	loads without crashing or leaking framework errors

tests/e2e/specs/quality/known-issues.spec.ts — 1 tests

#	Test
1	selecting "7 Days" must not display "8 Days"

tests/e2e/specs/quality/performance.spec.ts — 2 tests

#	Test
1	... loads within ...ms
2	dashboard stays interactive (no frozen UI) right after navigation

tests/e2e/specs/quality/rendering.spec.ts — 2 tests

#	Test
1	... renders meaningful content
2	... (...) loads without crashing

tests/e2e/specs/settings/permission-matrix.spec.ts — 3 tests

#	Test
1	renders the matrix with a MODULE column and role columns
2	lists module permission rows
3	shows the roles × modules summary

tests/e2e/specs/settings/profile.spec.ts — 4 tests

#	Test
1	shows the account details card with the user identity
2	lists access level, organization and login metadata
3	renders the recent activity feed
4	no leaked placeholder values

tests/e2e/specs/settings/roles.spec.ts — 5 tests

#	Test
1	exposes a Create New Role action
2	status tabs are present and switchable
3	negative: Next without a role name shows a required error
4	positive: a valid role name advances the wizard
5	Cancel wizard returns to the roles list

tests/e2e/specs/settings/theme-settings.spec.ts — 4 tests

#	Test
1	exposes Light and Dark editing modes
2	exposes import / export / reset / save controls
3	renders the editable colour token palette
4	switching to Dark Mode keeps the palette editable

tests/e2e/specs/settings/users.spec.ts — 10 tests

#	Test
1	shows summary stat cards
2	renders the users table with expected columns
3	status tabs switch the listing
4	search box filters the table
5	pagination controls are present
6	shows the three-step wizard (Identity → Resource Scope → Review)
7	negative: empty submit surfaces required-field errors
8	negative: invalid email is rejected
9	positive: a valid identity advances to the Resource Scope step
10	Cancel returns to the users list

tests/e2e/specs/settings/workflow.spec.ts — 3 tests

#	Test
1	renders the role hierarchy nodes
2	exposes the canvas zoom / fit controls
3	shows the reparent hint / quick actions

tests/e2e/specs/shell/navigation.spec.ts — 5 tests

#	Test
1	... > ... (...) loads
2	... stays stable after reload
3	hovering the rail expands it to reveal navigation labels
4	sidebar link navigates to Asset Management
5	logo returns to the dashboard from a deep page

tests/e2e/specs/shell/responsive.spec.ts — 1 tests

#	Test
1	... is usable on small viewports

tests/e2e/specs/shell/ui-consistency.spec.ts — 3 tests

#	Test
1	... has no horizontal overflow and consistent fonts
2	footer with version is shown on every integrated page
3	dark / light mode toggle changes the theme